NCP commutator-ideal refutation

THM-COMMUTATOR-KERNEL1 Proposition 1 / TASK-20260822-b69d29 / Lean 4.35.0-rc1

A = free associative Z-algebra on translations t_i and negation v

w = $v^2 - 1$; I = two-sided ideal generated by all xy – yx

Affine evaluation ρ

$\rho(v)^2 = 1$, so $\rho(w) = 0$

Thus w belongs to ker ρ.

Zero-generator augmentation ε

$\varepsilon(xy - yx) = 0$; $\varepsilon(w) = -1$

Thus w is outside I.

Therefore ker ρ is different from I for every B, n, and translation list s.